

AUST CSE LaTeX Template for CSE4250 Thesis Report

A thesis

Submitted in partial fulfillment of the requirements for the Degree of
Bachelor of Science in Computer Science and Engineering

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November 2024

CANDIDATES' DECLARATION

We, hereby, declare that the thesis presented in this report is the outcome of the investigation performed by us under the supervision of Name of the Supervisor, Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh. The work was spread over two final year courses, CSE4100: Project and Thesis I and CSE4250: Project and Thesis II, in accordance with the course curriculum of the Department for the Bachelor of Science in Computer Science and Engineering program.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere

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CERTIFICATION

This thesis titled, “AUST CSE LaTeX Template for CSE4250 Thesis Report”, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in November 2024.

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ACKNOWLEDGEMENT

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ABSTRACT

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Chapter 1

Introduction

1.1 Overview

Effective land administration and management are significant for a stable economy and a peaceful society. However, managing land resources remains one of the most challenging administrative tasks, particularly in developing countries such as Bangladesh. Here, traditional approaches to land management have faced obstacles that have been consistently characterized by inefficiency, corruption, and lack of transparency. These persistent issues result in significant consequences, including unclear property rights, hindered economic growth, and widespread public distrust.

Blockchain technology, widely recognized for its role in cryptocurrencies, has also gained attention for its potential to transform various administrative sectors, including land management. It is essentially a secure digital ledger capable of recording transactions on a network of computers. This decentralized approach ensures that each transaction is verified by multiple participants, making the recorded information permanent, transparent, and tamper-proof. Due to these inherent features, blockchain provides a promising solution to the fraud, corruption, and data manipulation issues prevalent in conventional administrative systems.

Bangladesh is currently experiencing rapid digital growth, driven by ongoing government efforts toward digital transformation. Consequently, exploring the role of blockchain technology in land administration and management is now a timely and relevant issue. By adopting blockchain, it could significantly streamline land registration and administration processes, minimize fraudulent practices, and improve trust between citizens and officials alike.

This study aims to explore the specific ways how blockchain technology can effectively solve existing problems within the land administration system in Bangladesh. It will identify

the key inefficiencies in the current system, investigate how blockchain can increase transparency and security, and evaluate the practical considerations and potential impacts of implementing such technology.

1.2 Problem Statement

Despite recent digitisation efforts, the land administration system in Bangladesh continues to suffer from multiple, interrelated problems that prevent reliable, transparent and equitable management of land records. The following points summarise these problems in detail.

- **Widespread corruption and fraudulent transactions.** Bribery, forged documents and unauthorised transfers remain common, undermining the integrity of land records and producing persistent legal disputes. High profile cases and international investigations underline the scale of illicit enrichment linked to the land sector. [10, 11, 13]
- **Large scale illegal occupation and encroachment.** Significant areas of public and private land are under illegal possession, complicating efforts to enforce rights and to manage land resources effectively. These occupations increase the administrative burden and raise the cost of conflict resolution. [11, 12]
- **Digitisation without verification.** Converting paper records into electronic files has improved access but often replicates existing errors and gaps because legacy records are not systematically verified during migration. As a result, digitisation alone has not removed opportunities for manipulation or restored confidence in recorded ownership. [15]
- **Weak data quality and record inconsistency.** Incomplete, contradictory or poorly indexed records reduce the usefulness of national land databases and make automated validation difficult. This poor data quality impedes search, reconciliation and reliable title verification across offices and regions.
- **Inadequate security and access controls.** Current digital systems often lack strong authentication, fine grained access controls and sufficient audit logging, which increases the risk of unauthorised changes and data leakage. Technical glitches and limited cybersecurity capacity further expose records to tampering.
- **Limited legal recognition of digital evidence.** Even when records are stored electronically, courts, banks and other institutions may hesitate to rely on them if legal frameworks do not clearly recognise the validity of digitally anchored records. This

legal uncertainty reduces the value of modernised records as collateral and as evidence in dispute resolution.

- **Operational dependence on manual procedures.** Many core processes still require paper based verification, physical signatures and in person approvals, which creates delays and sustains informal practices that enable corruption. Manual steps also increase transaction time and operational cost for citizens and administrators.
- **Poor service delivery in remote and low connectivity areas.** Offices outside urban centres face challenges in synchronising records, performing timely verification and participating fully in centralised systems. This digital divide exacerbates regional inequalities and leaves rural and marginal communities more vulnerable.
- **Vulnerability of marginalised and indigenous groups.** Customary and community land rights are often not well captured in formal record systems, leaving indigenous and other vulnerable populations exposed to dispossession and eviction with limited legal recourse. [14]
- **Limited integration with financial and legal systems.** Banks and courts require high assurance of record integrity before they accept land documents as collateral or evidence. When confidence is low, landholders face restricted access to formal credit and prolonged dispute resolution processes.
- **Lack of tested technology integration and scale pilots.** There is limited research and few validated pilot projects that demonstrate how advanced technologies such as blockchain, secure off chain storage and identity management can be integrated with institutional workflows and legal frameworks in Bangladesh. This gap makes it difficult for policy makers to adopt technology driven reforms with confidence.
- **Capacity and governance gaps.** Successful modernisation requires trained personnel, clear governance arrangements, role based responsibilities and change management. Weak institutional capacity, unclear endorsement rules and resistance to procedural change hinder effective implementation.

Together these problems create a system in which recorded land information cannot be relied upon with confidence, transaction costs are high, dispute volumes remain elevated and vulnerable groups bear disproportionate risk. The study that follows addresses these interlocking problems by designing a technology oriented approach that targets data integrity, verifiable processes, controlled disclosure and institutional alignment.

1.3 Motivation

Land in Bangladesh is a foundational asset for families, businesses and the national economy. Current land administration procedures remain largely paper based and manual, producing long delays, opaque decision processes and repeated opportunities for fraud. High profile reports of forged documents, illegal land occupation by powerful actors and informal payments to speed up services have eroded public trust and increased legal disputes. These realities create urgent motivation to design a system that can provide stronger proof of ownership, faster transactions and clear audit trails.

Existing digital initiatives such as the e Mutation platform have improved some operational steps but often replicate legacy verification gaps when they simply convert paper records into electronic files. Without reliable mechanisms for immutable recording and verified identities, digitization alone cannot prevent manipulation or restore confidence. The project described in this report uses a permissioned blockchain architecture built on Hyperledger Fabric together with off chain storage on IPFS to directly address these verification and integrity weaknesses. By design, each critical action is recorded on chain, while large documents remain accessible via secure off chain references.

Automation through smart contracts is a central motivation for this work because it reduces human intervention at key workflow points such as registration, sale, lease and mortgage. Smart contracts enforce business rules and require dual endorsement by both the participant and an authorised land inspector before a state change is final, removing informal gatekeeping and making approvals auditable. Ownership tokens and receipt non fungible tokens represent rights and agreements in a machine verifiable way, enabling faster transfers and transparent evidence for courts, banks and citizens.

Privacy and practical deployment considerations also motivate the chosen approach. Sensitive documents are stored in encrypted private data collections and channel permissions limit visibility to authorised organisations, so that public verification of essential facts can coexist with individual privacy. The system supports event driven notifications and off chain indexing for fast queries, which helps courts, banks and survey offices work with near real time data without exposing personal details. To serve remote and low connectivity areas, the consortium design anticipates regional peer nodes and local indexing to allow offline data capture and later reconciliation with the main network.

Broader social and economic motivations include lowering transaction costs, improving access to credit and enabling better land use planning. Secure on chain records reduce title risk and make land parcels usable as credible collateral, which can expand access to mortgage finance and boost productive investment. Transparent histories of ownership and encumbrance simplify dispute resolution and reduce the burden on courts and administration,

producing direct time and cost savings for citizens and the state.

Finally, institutional readiness and legal recognition are core practical motivations. This project focuses on a consortium governance model that invites the land ministry, banks and authorised agencies to operate nodes and define endorsement policies, creating a pathway for incremental policy alignment and pilot validation. The technical choices in this project aim not only to demonstrate a secure and transparent system but also to provide a realistic roadmap for policy makers, inspectors and community stakeholders to adopt the technology in a phased and legally compliant manner.

1.4 Objectives

The purpose of this research is to design, implement and assess a practical blockchain based land administration system that addresses persistent weaknesses in the current processes of Bangladesh. The study aims to produce a technically sound system while also offering a feasible roadmap for institutional adoption, stakeholder engagement and measurable operational improvement. The objectives below state a primary aim followed by specific implementation oriented goals.

Primary objective: Develop a secure, transparent and practical land administration framework using a permissioned blockchain architecture together with off chain storage and automation mechanisms that reduce fraud, accelerate common transactions and strengthen public trust in land records.

Specific objectives:

1. Design an end to end architecture based on a permissioned blockchain such as Hyperledger Fabric, integrated with off chain document storage like IPFS and privacy preserving data sharing to ensure immutability, verifiability and controlled access to sensitive records.
2. Implement smart contract driven workflows for core land processes including registration, ownership transfer, lease, mortgage and dispute logging so that business rules are enforced automatically and approvals become auditable.
3. Define identity and access management provisions for participant roles including land inspectors, registrars, banks, surveyors and citizens, and specify endorsement policies that reflect institutional responsibilities.
4. Provide mechanisms for document anchoring and indexing so that large legal documents remain accessible off chain while cryptographic references on chain provide tamper proof evidence of original records.

5. Incorporate selective disclosure and encrypted private data collections to preserve confidentiality while enabling public verification of essential facts.
6. Prepare a pilot deployment and test plan to evaluate system behaviour under realistic operational conditions, including performance, scalability, fault tolerance and data reconciliation for low connectivity environments.
7. Examine legal and institutional requirements for adoption and prepare policy oriented recommendations that enable phased integration with existing land administration processes.
8. Assess usability and stakeholder acceptance through user centred testing and refine workflows to reduce operational complexity and improve accessibility for diverse participants.

Practical goals and indicators: The project will be guided by practical goals that reflect technical robustness, operational usefulness and institutional readiness. Technical robustness includes verifiable records on chain, reliable off chain references and predictable performance as data volumes increase. Operational usefulness covers reduced processing time for common workflows, fewer manual interventions and improved ability for banks and courts to rely on recorded evidence. Institutional readiness refers to clear governance models, endorsement policies and documentation that support pilot adoption by government and partner organisations.

Indicators of impact:

- Integrity and auditability: Critical state changes and document anchors are cryptographically verifiable and produce clear audit trails.
- Performance and scalability: Transaction throughput and latency meet practical thresholds for pilot use and the architecture demonstrates paths for scaling the consortium network.
- Privacy and compliance: Private data controls and encryption meet baseline legal and stakeholder privacy requirements.
- Operational efficiency: Measurable reduction in average processing time for selected workflows and reduction in manual approvals compared to the baseline.
- Socio economic outcomes: Evidence that improved record reliability reduces title risk and facilitates access to formal credit for landowners.
- Adoption readiness: Existence of a governance model, policy brief and pilot results that together create a feasible path for phased institutional adoption.

Expected outputs:

- A working prototype implementing the core blockchain network, smart contracts and off chain storage integration suitable for pilot testing.
- Technical documentation including architecture diagrams, data models, smart contract specifications and deployment instructions to assist replication.
- An assessment report containing quantitative performance data, security observations and findings from user acceptance testing in pilot scenarios.
- Policy oriented notes that outline recommended endorsement policies, stakeholder roles and steps for legal recognition and phased rollout.

By meeting these objectives the research intends to deliver a concrete technical solution and an evidence based pathway that policy makers, administrators and development partners can use to pilot and scale a blockchain based land administration system in Bangladesh. The focus is on producing practical improvements in trust, efficiency and access to services while identifying realistic constraints and next steps for wider adoption.

1.5 Organization of the Book

An overview of how the book is structured and what each chapter contains.

Chapter 2

Background Study and Literature Review

2.1 Overview

Land administration plays a crucial role in establishing secure property rights, reducing disputes, and supporting socio-economic development. However, conventional land governance systems in many countries, particularly in developing regions like Bangladesh, continue to face challenges related to inefficiency, corruption, lack of transparency, and unreliable record-keeping. These limitations have often resulted in overlapping ownership claims, unauthorized land transactions, and a general lack of public trust in land management authorities.

In recent years, researchers and policymakers have turned their attention to emerging technologies such as blockchain to address these long-standing challenges. Blockchain technology, known for its decentralization, immutability, and transparency, has been widely recognized as a promising solution for improving the integrity and efficiency of land administration systems. The integration of smart contracts and decentralized storage further enhances automation, security, and accessibility in land record management.

The reviewed literature reveals several key themes and trends in this field. Existing research demonstrates that blockchain-based land administration can significantly reduce fraudulent activities, automate ownership transfers, and provide tamper-proof records of property ownership. Studies have also highlighted the importance of stakeholder involvement, decentralized document storage through technologies like InterPlanetary File System, and the potential for smart contracts to eliminate manual errors and delays in land transactions.

Despite these advancements, the literature also identifies persistent gaps. Most existing works focus on prototype-level systems, with limited attention given to large-scale deployment, legal integration, system scalability, and socio-political challenges. Addressing these gaps is critical to fully realizing the potential of blockchain in transforming land governance,

particularly in developing countries where these challenges are most prevalent.

2.2 Background Study

Land administration forms the foundation of property rights, economic development, and social stability in any nation. However, the historical management of land records, especially in developing countries like Bangladesh, has been marked by significant challenges, including inefficient processes, corruption, and lack of transparency [10, 11]. Traditionally, land ownership records in Bangladesh have been maintained through centralized, paper-based systems vulnerable to errors, manipulation, and unauthorized alterations [12]. This fragile system has contributed to an environment where disputes over land ownership, illegal occupations, and overlapping claims are frequent, leading to considerable socio-economic instability [13, 14].

Multiple reports have highlighted how deep-rooted governance challenges in Bangladesh's land administration continue to erode public trust. Investigations have revealed that fraudulent practices, political interference, and a lack of accountability often allow influential groups to manipulate records or acquire land through illicit means, leaving genuine owners and marginalized communities vulnerable [13, 14]. These problems are not unique to Bangladesh. Across many developing regions, weak land governance structures have led to similar concerns, undermining property rights, discouraging investment, and impeding socio-economic progress [15].

In response to these issues, governments have initiated efforts to modernize land administration by digitizing land records and introducing information technology systems [15]. Bangladesh has implemented measures to phase out manual documentation in land offices, aiming to improve record accessibility and administrative efficiency [15]. Nevertheless, experts agree that digitized centralized systems alone cannot fully resolve the inherent challenges of data tampering, lack of transparency, or the risk of single points of failure [16]. These shortcomings have driven global interest in exploring more robust technological frameworks capable of providing secure, transparent, and tamper-proof land governance.

Blockchain technology has emerged as a promising solution to these enduring challenges. Introduced in 2008 with Bitcoin, blockchain represents a decentralized ledger system that allows transactions to be securely recorded, cryptographically verified, and distributed across multiple nodes, ensuring data integrity without reliance on a central authority [1]. In the context of land administration, blockchain provides immutability, traceability, and transparency, making unauthorized alterations or fraudulent modifications virtually impossible [9].

Moreover, the integration of smart contracts within blockchain systems enhances efficiency by automating land transactions based on predefined conditions. This eliminates the need for excessive manual verification, reduces processing time, and minimizes human errors [7]. Recent developments such as the Blockchain-enabled Land Records System (BLRS) have demonstrated how blockchain and smart contracts can be effectively applied to land administration. The BLRS model employs Ethereum blockchain technology, combined with smart contracts, to facilitate secure land transfers, track ownership history, manage land disputes, and enable transparent government oversight [3].

In addition to blockchain's distributed ledger capabilities, supporting technologies like the InterPlanetary File System (IPFS) have been integrated to securely store large land-related documents off-chain while preserving cryptographic links to the blockchain records [2]. This enhances system scalability and ensures that both ownership data and supporting documentation remain tamper-resistant and easily accessible.

Globally, countries such as India, Pakistan, and several African nations have also explored blockchain-based land registry systems to strengthen property rights, reduce corruption, and improve efficiency [4]. Studies show that these systems offer the potential to significantly reduce fraudulent land transactions, enhance public trust, and foster a more reliable and transparent land governance framework [3, 9].

2.3 Literature Review

Ref No.	Technology	Key Feature	Limitation
[1]	Ethereum, Hybrid Blockchain, Smart Contracts	Phase-wise Blockchain adoption, transparency, immutable land records	Scalability concerns, early-stage adoption challenges
[2]	Blockchain, IPFS, Ethereum	Immutable land records, decentralized application	Depends on public Blockchain, scalability limits
[3]	Blockchain	Automated land records, secure transactions, smart contracts	No real-time system validation discussed
[4]	Blockchain, Smart Contracts	Transparent land ownership, reduced fraud risk	Requires political will, stakeholder resistance possible
[5]	Ethereum, Smart Contracts, Distributed Ledger	Prevents double-spending, enhances transparency in lease & mortgage	Does not address all Blockchain limitations fully
[6]	Blockchain, Smart Contracts, AI, IoT	Multi-layer system, real-time monitoring, AI for fraud detection	Regulatory challenges, scalability concerns
[7]	Blockchain, Smart Contracts, Decentralized Platform	Eliminates intermediaries, enhances transparency	Prototype stage, lacks real-world validation
[8]	Blockchain, AI, IPFS, Ethereum	AI for fraud detection, IPFS for secure data storage	High complexity, requires AI expertise
[9]	Blockchain, Smart Contracts, DApp	Tamper-resistant platform, decentralized registry	Focuses on concept, lacks large-scale testing

Table 2.1: Summary of Blockchain-Based Land Management Papers

The application of blockchain technology in land administration has attracted considerable research interest due to its potential to deliver secure, transparent, and tamper-resistant record-keeping. Across existing studies, blockchain is recognized for its ability to overcome persistent challenges in conventional land management, including manual errors, centralized control, and vulnerability to fraud.

A common approach adopted by researchers involves utilizing decentralized ledger technology to record land ownership, transactions, and related activities. Through distributed consensus mechanisms, blockchain eliminates the need for a central authority and enhances

transparency and auditability across land records [1,2,6,9]. Smart contracts, implemented primarily on Ethereum-based platforms, are widely integrated to automate processes such as ownership transfers, mortgage agreements, and land leasing. These self-executing contracts reduce reliance on intermediaries, minimize human errors, and provide a tamper-proof transaction history [2,3,5,9].

The use of off-chain storage mechanisms, particularly the InterPlanetary File System (IPFS), has been highlighted in several studies to address blockchain's storage limitations. IPFS facilitates secure, decentralized storage for large documents like land deeds and verification reports, while maintaining verifiable cryptographic links to on-chain records. This hybrid approach enhances data accessibility and system scalability without compromising integrity [2,8].

AI integration has also emerged as a key enhancement to blockchain-based land management frameworks. AI models are applied for fraud detection, anomaly classification, and predictive insights, significantly improving the security and reliability of land registry platforms [6,8]. Additionally, some frameworks explore incorporating real-time monitoring capabilities through IoT devices, enabling continuous tracking of land use and preventing unauthorized activities [6].

Stakeholder inclusion within blockchain ecosystems is identified as critical for building trust and ensuring operational transparency. Systems designed to involve government authorities, landowners, buyers, legal representatives, and inspectors promote authenticated transactions and reduce the potential for disputes [1,6,9]. Furthermore, certain studies propose advanced features such as land classification, ownership history tracking, and automated dispute resolution mechanisms to improve administrative efficiency [3,6,9].

Despite these advancements, significant limitations and barriers remain. Scalability constraints, especially in public blockchain environments, are a recurring concern [2,6]. Many proposed frameworks remain at the prototype stage, lacking comprehensive real-world validation or integration with existing land administration systems [7,9]. Regulatory challenges, legal recognition of blockchain-based land records, and alignment with institutional policies continue to hinder large-scale deployment [4,5].

Moreover, socio-political readiness and public trust are essential factors influencing the success of blockchain-driven land reforms. While technical solutions provide security and automation, effective implementation depends on addressing governance frameworks, regulatory alignment, and stakeholder education to facilitate adoption [4,5].

In conclusion, the reviewed studies demonstrate the promising role of blockchain in transforming land administration through decentralization, immutability, and automation. Integrating AI, IPFS, and IoT further strengthens system capabilities. However, practical challenges related to scalability, legal frameworks, and institutional readiness highlight the

need for continued research, stakeholder engagement, and pilot deployments to achieve widespread adoption of blockchain-based land management solutions.

2.4 Gap Analysis

The literature on blockchain-based land administration demonstrates significant progress in conceptual frameworks, prototype development, and the technical evaluation of decentralized applications for secure land registration. Most existing studies have successfully demonstrated the potential of blockchain technology to enhance transparency, reduce fraudulent activities, and automate ownership transfers through smart contracts. Furthermore, many of the reviewed works integrate additional technologies such as InterPlanetary File System for decentralized storage and emphasize stakeholder involvement to improve trust and accountability in land governance.

Despite these advancements, critical gaps remain that warrant further investigation. The majority of studies are primarily focused on technical implementations at the prototype level, while large-scale real-world deployment, integration with existing legal frameworks, and socio-political acceptance have not been adequately explored. Additionally, scalability, interoperability, and the regulatory challenges associated with blockchain adoption in land management are areas that require deeper research attention. The key gaps identified from the reviewed literature are summarized below:

- Integration of blockchain-based systems with existing centralized land records and legal infrastructure remains unexplored in many studies.
- Socio-political challenges such as government readiness, institutional capacity, and stakeholder resistance to adopting blockchain-based land governance have received limited attention.
- The scalability of blockchain networks, especially under high transaction volumes common in national land registries, is insufficiently addressed.
- There is limited comparison of performance across different consensus models for land management, most studies focus only on a single consensus model. So which model performs better in different situation for land management and administration remains unexplored.
- The role of emerging technologies such as Artificial Intelligence for fraud detection, automated decision-making, or predictive analytics in land administration systems remains conceptual with minimal practical implementation.

- User accessibility, digital literacy barriers, and the socio-economic inclusion of marginalized communities within blockchain-based land governance have not been fully investigated.
- Most existing studies are limited to prototype-level implementations and lack large-scale field deployment in real-world land administration environments.

These gaps indicate that while blockchain technology holds significant promise for revolutionizing land administration, comprehensive research is required to address technical, legal, and social complexities to enable effective and sustainable implementation at scale.

2.5 Summary

This chapter has presented the historical background and recent developments in the field of blockchain-based land administration. The literature demonstrates that blockchain, smart contracts, and decentralized storage systems have the potential to enhance transparency, prevent unauthorized alterations, and automate land transactions. However, several critical gaps remain. Most studies are limited to small-scale prototypes, while challenges such as legal integration, scalability, and socio-political readiness have not been sufficiently addressed.

To overcome these limitations, future research must focus on developing practical, scalable, and legally compatible solutions that can function within the socio-political landscape of countries like Bangladesh. Our proposed project aims to address these challenges by designing a secure and transparent land administration system using blockchain technology, integrated with smart contracts and decentralized storage mechanisms.

Unlike previous studies, our project will emphasize real-world applicability by focusing on stakeholder involvement, ease of system accessibility, and alignment with existing land governance structures. Additionally, features such as automated ownership verification, role-based access control, and integration of off-chain storage through IPFS will help ensure both technical efficiency and data integrity. By addressing these gaps, our project can contribute to establishing a more reliable, transparent, and scalable land administration framework.

Chapter 3

Methodology

3.1 Introduction/Overview

An overview of the methodology employed in this study. Discuss why this methodology is appropriate for addressing your research questions.

3.2 Dataset

Details regarding the dataset used for analysis. Describe its origin, size, and relevance to your research objectives.

3.2.1 Data Selection

Criteria for selecting data relevant to this study. Explain how you determined which data points were included.

3.2.2 Data Preprocessing

Steps taken to preprocess data before analysis. Discuss any cleaning, normalization, or transformation processes applied to the dataset.

3.3 Proposed Methodology and Design

This section describes the proposed methodology and the system design in detail. It presents the overall system structure and then discusses the principal workflows used for registration, transaction processing (buy/sell), and lease/mortgage operations. Each workflow is accompanied by a diagram and a focused description to explain the components involved, the sequence of actions, and the information exchanged.

3.3.1 Overall End-to-End Flow

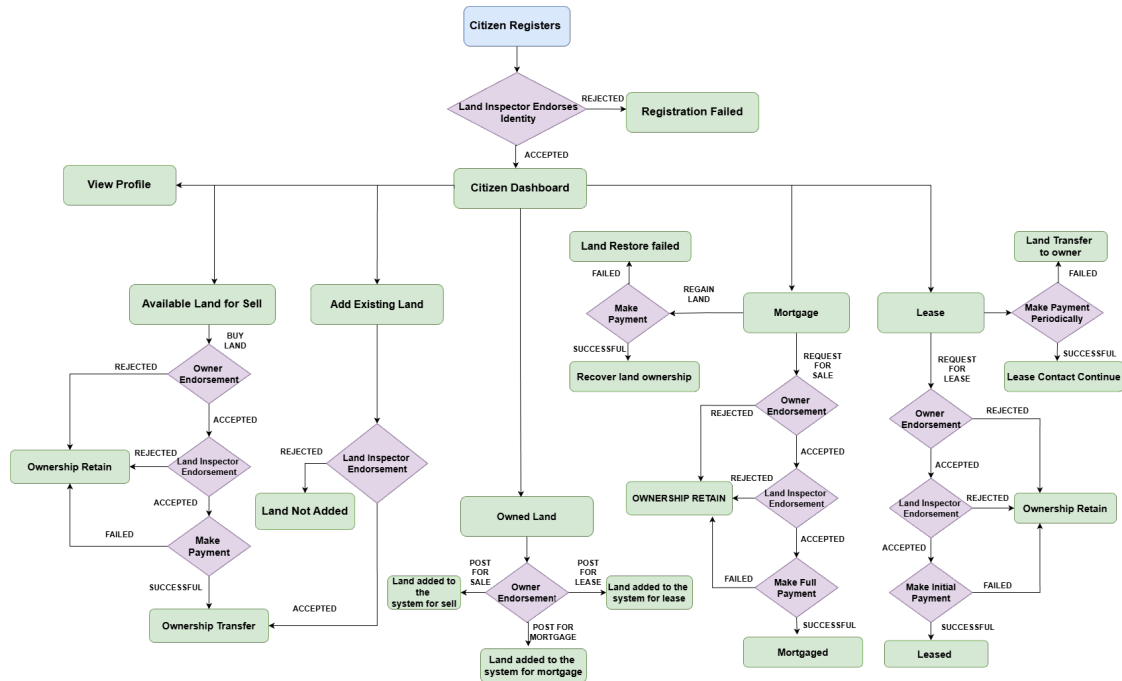


Figure 3.1: End-to-end system flow.

The foregoing diagram illustrates the end-to-end architecture. The system is organised in logical layers: (i) client-facing applications that provide forms, document upload and user interaction; (ii) an orchestration/API layer that validates requests, coordinates multi-step processes and acts as the gateway to backend services; (iii) a ledger layer that records canonical transaction metadata and enforces business rules through smart contract logic; and (iv) supporting subsystems including off-chain storage for large documents, privacy controls for restricted data sharing, identity and certificate services, and monitoring/operations tooling.

Data flow proceeds from user action through the API layer to the ledger. Large documents are stored off-chain and referenced on the ledger by compact identifiers. Policy boundaries and trust domains are made explicit in the diagram: they indicate which data is publicly visible, which data is restricted to specific participants, and where cryptographic checks are applied to guarantee integrity and non-repudiation.

3.3.2 Registration and Inspector Approval

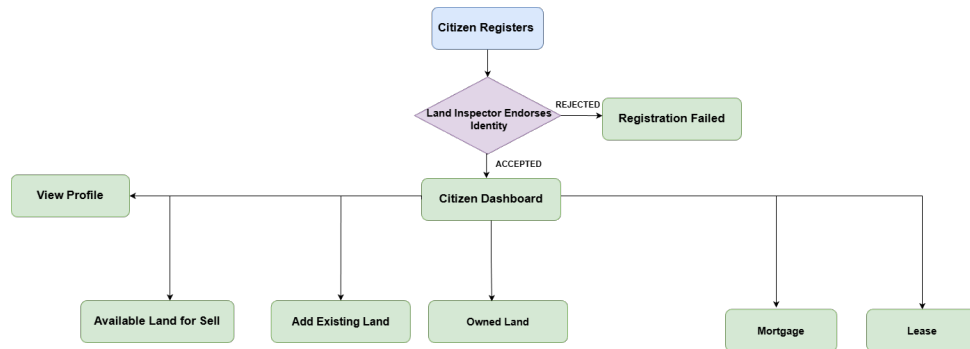


Figure 3.2: Registration and approval.

The registration workflow begins with a prospective owner submitting identity proofs and property documents via the client interface. Submitted documents are optionally encrypted at the client and stored in the off-chain store; a content reference (hash/identifier) is returned and included in the registration request. The API layer validates the request schema and forwards a signed proposal to the designated endorsing peers.

An inspector role is introduced to perform manual or semi-automated checks. The inspector reviews uploaded evidence, verifies spatial and legal attributes, and records a verification outcome. If the inspector approves, the transaction proposal obtains the required endorsements and is submitted to the ordering service for inclusion into the ledger. The ledger records the canonical registration record and the associated document references; application-level indices are updated and stakeholders are notified. Any rejection by the inspector triggers a documented remedial process (resubmission, correction or dispute initiation), and the attempt is logged for audit.

3.3.3 Buy–Sell Workflow

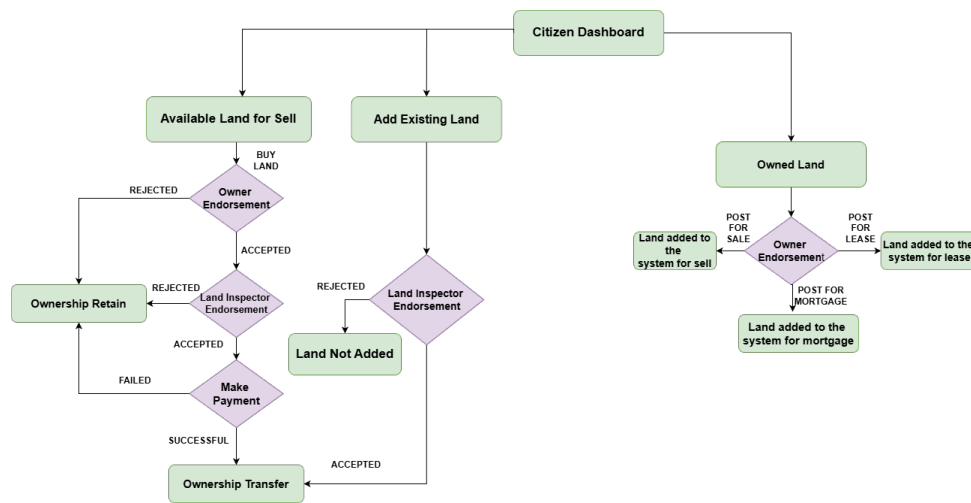


Figure 3.3: Buy–sell transaction flow.

The buy/sell workflow covers listing a property, receiving offers, and transferring ownership. A seller creates a listing containing parcel metadata and reference to supporting documents. Buyers submit offers through the client; offers may include payment intent or require bank approval. The API orchestrates multi-party interactions: it validates offers, coordinates escrow or bank interactions (if payment guarantees are required), and prepares a transaction proposal that encodes the transfer conditions.

Endorsement policies enforce that required parties (seller, buyer, relevant authority, and optionally an inspector or bank) sign off on the transfer. Upon satisfying endorsement and settlement conditions, the ordering service sequences the transaction and peers commit the change of ownership. The ledger stores the transfer record and a new deed reference; post-commit actions include releasing escrow funds, notifying registries and updating public indices. Any conditional clauses (e.g., contingent on mortgage clearance) are represented in the contract logic and enforced before final commit.

3.3.4 Lease and Mortgage Workflow

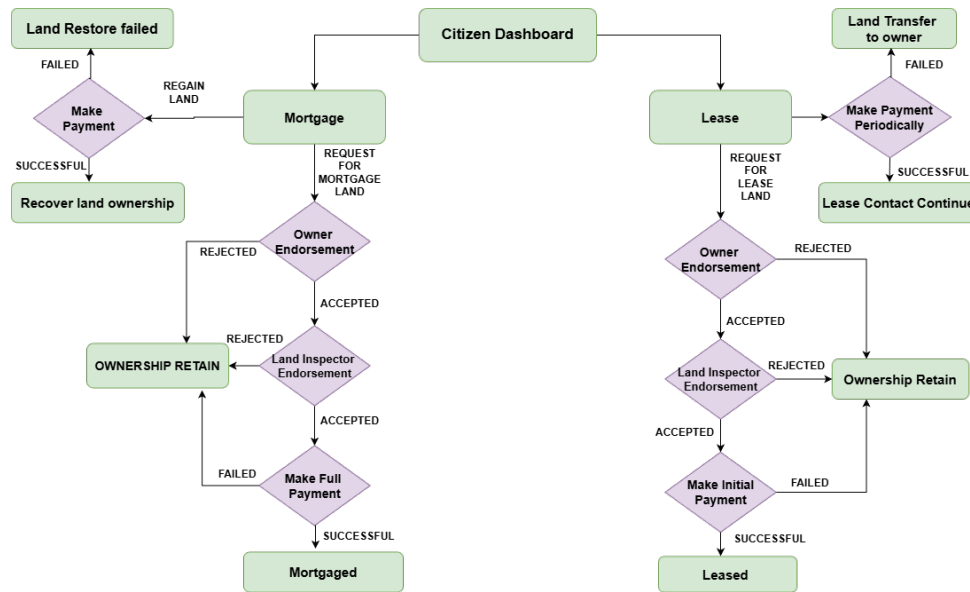


Figure 3.4: Lease and mortgage operations.

Lease and mortgage processes involve extended-term obligations and additional private data. A lease initiation records contractual terms (duration, rent schedule, responsibilities) and references any attached documents. Lease modifications and terminations follow defined contract methods and require authorised signatures. For mortgage operations, financial parties (lenders) participate as stakeholders: sensitive loan terms and collateral details are held in privacy-preserving collections accessible only to authorised participants. The workflow includes loan application, valuation/inspection, lender approval, and registration of a lien against the parcel record.

Repayment events, escrow management and release-of-lien actions are emitted as ledger events and cause updates to the parcel's status. Foreclosure or release actions are subject to policy-enforced procedures and may require multi-party endorsement.

3.3.5 Component Responsibilities

To support the workflows above, components expose clearly defined interfaces:

- **Client:** input validation, secure document upload and digital signing of proposals.
- **API/Orchestrator:** request validation, session/auth management, workflow coordination and event emission.
- **Ledger (smart contracts):** domain logic implementation, state transitions, endorsement checks and event generation.

- **Off-chain store:** reliable storage for large artifacts with content-addressable identifiers and retention policies.
- **Identity service:** credential issuance, attribute management and revocation.
- **Monitoring and logging:** operational telemetry, alerting and audit trails.

Each interface is documented with input/output schemas, expected error conditions and test vectors.

3.3.6 Transaction Lifecycle and Dataflows

The system follows a structured and repeatable transaction lifecycle. Each step is well-defined to ensure correctness, reproducibility and efficient tracing. The complete lifecycle is outlined below:

- **Client Preparation:** The user gathers required metadata and documents; client-side validations are performed before submission.
- **Off-chain Document Persistence:** Large files are uploaded to the off-chain store. A content reference is generated and included in the transaction request.
- **Proposal Construction:** The client constructs a digitally signed transaction proposal containing metadata, document references and role credentials.
- **Endorsement Phase:** Designated peers simulate the transaction, validate its structure and verify access rights according to configured policies.
- **Ordering and Block Formation:** Endorsed proposals are forwarded to the ordering service, which batches them into blocks and determines final sequence.
- **Validation and Commit:** Network peers validate endorsements, check versioning constraints and commit the transaction to their local ledgers.
- **Event Handling and Post-Processing:** After commit, system events trigger index updates, notifications, auxiliary workflows and external system integrations.
- **Audit and Traceability:** Each step produces trace logs and metadata to support audit, debugging and performance evaluation.

3.3.7 Security and Privacy Considerations

Security and privacy mechanisms are incorporated throughout the system to ensure confidentiality, integrity and controlled access. The major considerations include:

- **Identity and Authentication:** All participants authenticate using certificate-based identities. Roles and permissions are encoded within issued credentials to enforce consistent access control.
- **Data Integrity and Non-repudiation:** All submitted data—including off-chain documents—is referenced through cryptographic hashes. Signed transactions guarantee authenticity and prevent tampering.
- **Privacy-aware Data Segregation:** Sensitive records are kept within restricted-access data partitions. Only authorised peers can view confidential details, ensuring compliance with privacy policies.
- **Secure Storage and Key Management:** Private keys are stored in secure key vaults or protected client environments. Regular rotation and revocation procedures reduce long-term exposure risks.
- **Auditability and Monitoring:** Comprehensive logs and system events provide a tamper-evident audit trail. Monitoring tools track abnormal patterns and support early detection of potential breaches.

3.3.8 Validation and Evaluation

The system design is evaluated through structured validation activities that test correctness, robustness and performance. The evaluation strategy consists of:

- **Functional Testing:** Unit and integration tests verify the correctness of core modules, including registration, transfer, lease and mortgage workflows. Role-based acceptance tests ensure proper enforcement of approval rules.
- **Performance Benchmarking:** Throughput, latency and resource consumption are measured under different network loads, endorsement policies and transaction volumes. These tests help determine scalability limits.
- **Resilience and Fault Tolerance:** Controlled failure scenarios—such as peer outages, ordering-node failures or network partitions—are simulated to examine system behaviour and recovery capability.

- **Storage and Data Growth Analysis:** The growth of the ledger and off-chain documents is monitored to evaluate long-term storage requirements and indexing performance.
- **Privacy Compliance Checks:** Tests ensure that sensitive information remains restricted to authorised entities and is not exposed during normal operation or fault conditions.
- **Reproducibility:** All tests are supported by benchmark scripts and documented procedures to ensure that results can be reproduced consistently across environments.

3.4 Implementation

Discussion on how the proposed methodology was implemented. Describe any tools or technologies used during implementation.

3.5 Summary

A summary of key findings from the methodology chapter. Highlight any significant insights gained from your methodological approach.

Chapter 4

Experiments and Results Analysis

4.1 Introduction/Overview

An overview of experiments conducted and results obtained. Briefly describe what was tested and why it matters.

4.2 Modern Tools

Description of modern tools utilized during experiments. Discuss their relevance and effectiveness in achieving your research objectives.

4.3 Result Analysis

Analysis of results obtained from experiments conducted. Present findings clearly, using tables or figures where appropriate.

4.4 Impact Analysis and Sustainability

Discussion on the impact of findings and sustainability considerations. Evaluate how your results contribute to long-term solutions or practices.

4.5 Summary

A summary encapsulating key points discussed in this chapter. Reinforce why these results are significant for future research or practice.

Chapter 5

Project Management and Cost Analysis

5.1 Project Management

An overview of project management strategies used in this study. Discuss planning, execution, monitoring, and control aspects.

5.2 Cost Analysis

Table 5.1: Estimated project costs (BDT)

Cost category	Type	Estimate (BDT)
Smart-contract development	One-time	3,037,500
Front-end & Dashboard UI/UX	One-time	1,822,500
Back-end & Gateway API	One-time	1,458,000
Blockchain network infrastructure (PoA nodes)	Recurring (annual)	607,500 / yr
IPFS storage & pinning	Recurring (annual)	486,000 / yr
Security audit & penetration testing	One-time	364,500
Off-chain indexing & search service	One-time + Recurring	243,000 (one-time) + 607,500 / yr
Maintenance & support	Recurring (annual)	1,215,000 / yr
Training & onboarding	One-time	364,500
Total one-time development costs		7,290,000
Total recurring annual costs		2,916,000 / yr

Description

Smart-contract development — 3,037,500 BDT (one-time): Chaincode for registration, transfer, lease, mortgage and audit modules; unit and integration testing; code review and deployment automation.

Front-end & dashboard (UI/UX) — 1,822,500 BDT (one-time): User interfaces for citizens, inspectors and administrators; responsive design, usability testing and UI iterations.

Back-end & gateway API — 1,458,000 BDT (one-time): API gateway, workflow orchestration, authentication/authorization, Fabric SDK integration and connectors to off-chain services.

Blockchain infrastructure (PoA nodes) — 607,500 BDT (recurring/yr): Annual hosting, monitoring and operational costs for ordering and peer nodes supporting the permissioned network.

IPFS storage & pinning — 486,000 BDT (recurring/yr): Annual cost for off-chain document storage, pinning guarantees and basic retrieval service.

Security audit & penetration testing — 364,500 BDT (one-time): External security review of smart contracts and APIs, plus penetration testing engagement.

Off-chain indexing — setup — 243,000 BDT (one-time): Initial design and implementation of the event-to-index pipeline and search configuration.

Off-chain indexing — hosting — 607,500 BDT (recurring/yr): Recurring hosting, maintenance and scaling costs for the search/index service.

Maintenance & support — 1,215,000 BDT (recurring/yr): Ongoing developer/operations retainer covering patches, minor feature work and SLA-based support.

Training & onboarding — 364,500 BDT (one-time): Pilot-region training sessions, materials, and initial user onboarding support.

5.3 Project Scheduling

The project follows a structured twelve-month schedule that divides major development activities into well-defined phases. The timeline ensures progressive refinement of system components, parallel development where appropriate, and adequate time for testing, validation, and documentation. Figure 5.1 illustrates the overall schedule for the major tasks across the one-year implementation period.

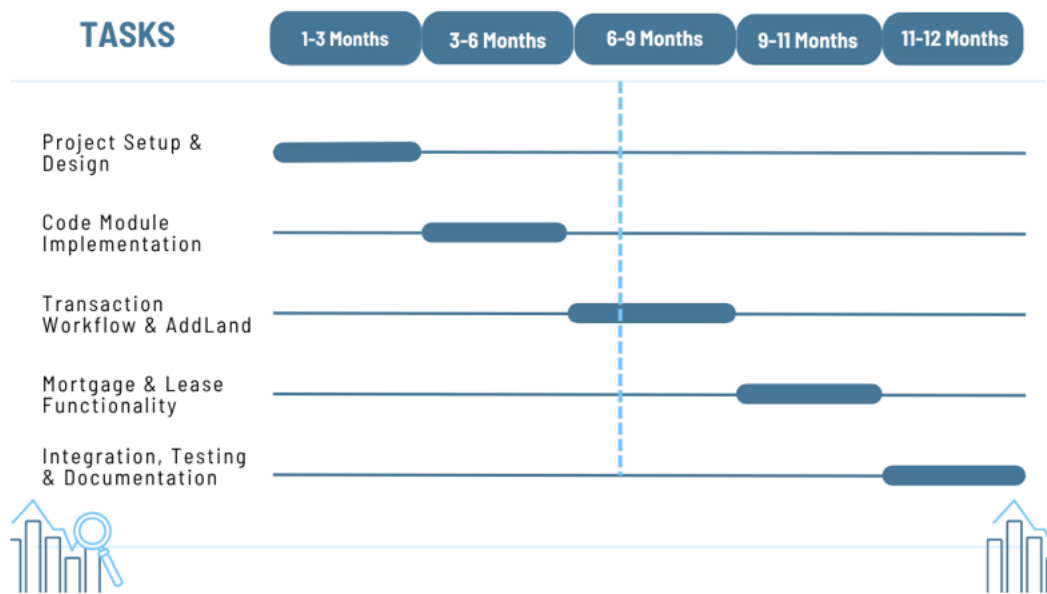


Figure 5.1: Project schedule across the twelve-month development cycle

1. Project Setup and Design (Months 1–3)

This phase establishes the foundation of the entire system. The key activities include:

- Initial requirement analysis and stakeholder alignment.
- High-level system architecture design, including ledger design, off-chain storage plan, identity model, and workflow structure.
- Environment setup for development, repository configuration, and CI/CD pipeline preparation.

This early-stage preparation ensures that all subsequent development tasks proceed within a stable and well-defined technical framework.

2. Code Module Implementation (Months 3–6)

During this period, the core modules of the platform are implemented. Major work includes:

- Development of foundational smart-contract components.
- Implementation of the gateway API and supporting back-end services.
- Construction of user interface elements and administrative dashboards.

These modules form the building blocks required for executing various land-related transactions.

3. Transaction Workflow and AddLand Features (Months 6–9)

This phase focuses on building and integrating the transaction logic that governs system behaviour. Key activities include:

- Implementation of end-to-end workflows for registration, land addition, and ownership transfer.
- Integration of ledger events with back-end services and off-chain indexing.
- Validation of transaction flows through internal testing and scenario-driven simulations.

This stage ensures that the system can securely and transparently support core land administration functions.

4. Mortgage and Lease Functionality (Months 9–11)

After establishing the primary workflows, the extended financial functionalities are developed:

- Implementation of lease request and approval processes.
- Mortgage creation, verification, and release workflows.
- Additional permission and validation steps required for financial institutions or inspectors.

These features complete the breadth of operations expected in a comprehensive land administration platform.

5. Integration, Testing, and Documentation (Months 11–12)

The final stage consolidates all components and prepares the system for deployment:

- System-wide integration and interoperability testing.
- Formal security review and performance validation.

- Preparation of user manuals, technical documentation, and training materials.
- Final refinements based on pilot testing feedback.

This ensures that the system is operationally stable, well-documented, and ready for rollout.

Overall, the project scheduling structure supports a logical progression from foundational design to functional development and final verification. By distributing work across overlapping but coordinated phases, the timeline aligns both development efficiency and system reliability throughout the one-year implementation cycle.

Chapter 6

Ethics and Professional Responsibilities

6.1 Introduction/Overview

An overview of ethical considerations in engineering practices. Discuss why ethics are crucial in your field of study.

6.2 Identify and Apply Ethical and Professional Responsibilities

Discussion on identifying and applying ethical responsibilities in engineering projects. Provide examples where applicable.

Chapter 7

Identification of Complex Engineering Problems and Activities

7.1 Complex Engineering Problem

7.1.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 7.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 7.1: Mapping with complex problem solving.

P1	P2	P3	P4	P5	P6	P7
Depth of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Inter-dependence
✓	✓					

7.1.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 7.2).

Table 7.2: Mapping with complex engineering activities.

A1	A2	A3	A4	A5
Range of Sources	Level of Interaction	Innovation	Consequences for Society and Environment	Familiarity
✓	✓			

Chapter 8

Conclusion and Future Works

Summarize key findings from your research, emphasizing their significance. Discuss potential future work that could build upon your findings or address remaining gaps.

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